

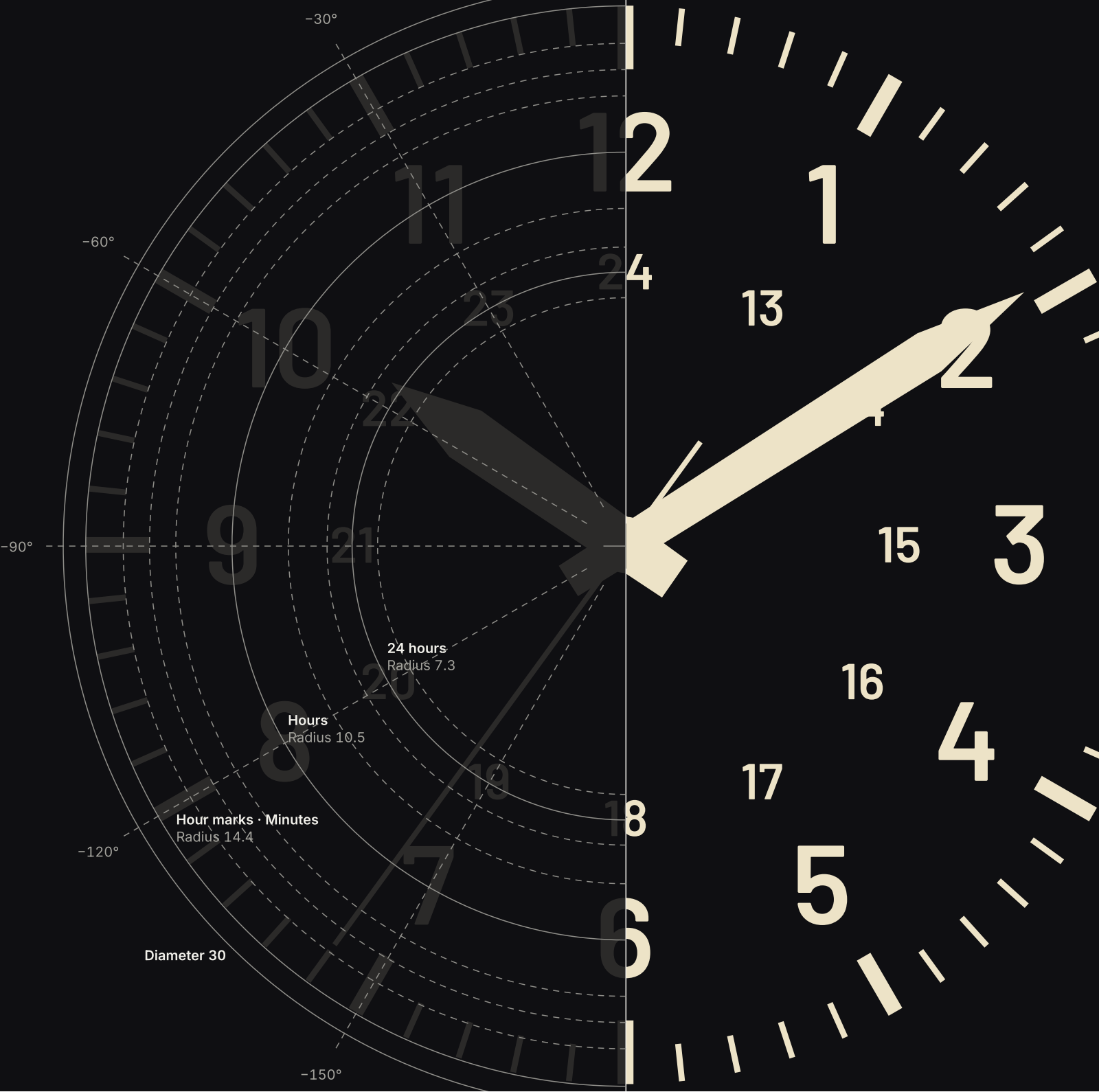
# Dial Tool

## Guidebook

beta 2.3 · 4 Oct 2026

A dial is a short list of rings. Left of the line is the list: every radius and angle the settings name. Right of it is what the list prints.

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# Start

One Python file that runs inside DrawBot, a free drawing app for macOS. It draws a dial as a list of rings, in millimetres.



Install

## Three steps, one file.

- 1 **DrawBot 3.132** or newer, from [drawbot.com](https://drawbot.com).
- 2 **dial.py** from [github.com/duztman/dial-tool](https://github.com/duztman/dial-tool). Branch main is the released version; dev is the newest work.
- 3 Open dial.py in DrawBot and press **⌘R**. The window opens.



Speed

## One optional package.

Knockouts and hollow shapes are cut by combining outlines. DrawBot's own routine is slow; skia-pathops is 2 to 14 times faster.

In DrawBot: Python → Install Python Packages → **skia-pathops** → Go. The Export log then opens with “shape combining: skia-pathops”.

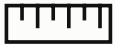


Memory

## Nothing to save.

Settings are stored as you work. Press **⌘R** again, or quit DrawBot: the last session reopens as you left it.

Export → Save... writes a named copy; Load... opens one. The nine dials in this book come as such files.



Unit

## mm

Every length is a millimetre, drawn at true size. Radii are measured from the dial's centre.



Angle

## $0^\circ = 12$

Clock angles, growing clockwise.  $90^\circ$  is 3 o'clock,  $180^\circ$  is 6,  $-90^\circ$  is 9.



Shape

## Fills

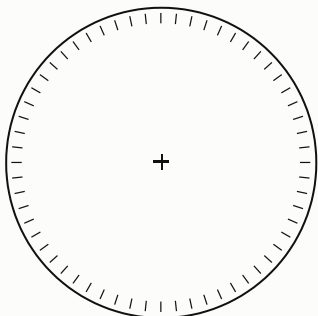
Everything exported is a filled outline, never a stroke. A 0.12 mm tick is 0.12 mm in Illustrator.



Order

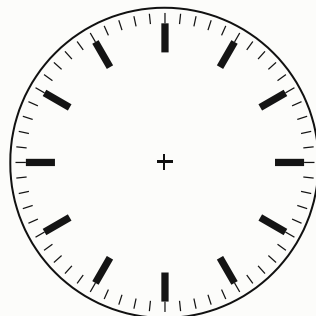
## Top first

The ring list is the stacking order. The top ring is drawn last, in front of the others.



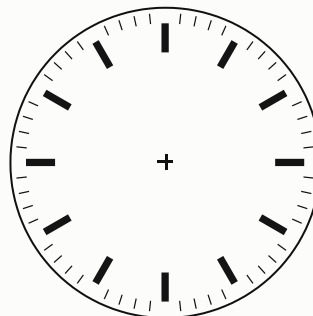
1 · ticks

Add ring... → ticks · Count 60



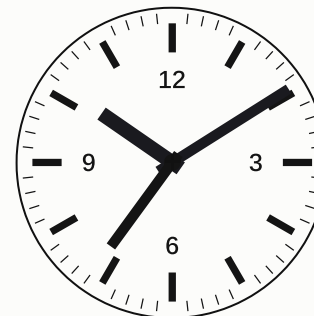
2 · markers

added above the ticks



3 · give way

ticks: Skip where markers above



4 · numerals, hands

Every 3 · Radius 8 · Size 2.4



Watch

It runs only inside DrawBot. The window, the fonts and the export all come from DrawBot's own Python.



Watch

The canvas preview needs macOS 14 or newer. Older systems get the PDF preview, and the log says why.

# The window

Settings on the left, the dial on the right, time underneath. Overleaf it is redrawn whole, in grey, from the tool's own measurements.

①

## Sections

Dial, Rings, Hands, Date, Export. One is shown at a time.

②

## Ring list

Top ring in front. The checkbox hides a ring; double-click a name to change it. The selected ring is the one the settings edit.

③

## Ring tools

Add ring... puts a new ring above the selected one. Duplicate, Delete, move up, move down.

④

## The ring's settings

One scrolling list. Rows that don't apply to the ring's kind are hidden, not greyed out.

⑤

## Preview

Follows every change. Pinch or ⌘-scroll zooms about the cursor; two fingers pan. With Guides on, the selected ring's edges are orange.

⑥

## Time

Drag, or type 10:09:36. Now sets the clock's time; Play runs the hands.

⑦

## Guides, Canvas · PDF, Fit

Guides are never exported. Canvas is the fast preview; PDF is DrawBot's own viewer, the reference. Fit shows the whole page.

⑧

## Status

Ring, font, diameter; then how long the geometry (build) and the drawing (frame) took.



Number

**Slider, field and stepper show one value.**

Length

Width · Ø

Drag for a rough value, type for an exact one, step for the last hundredth. Typing past a slider's end extends the slider: the ranges are habits, not limits.



Typing

**Fields do arithmetic.**

|      |          |
|------|----------|
| 1.2  | sets 1.2 |
| +0.5 | adds 0.5 |
| x0.5 | halves   |
| /3   | a third  |

Useful when a drawing says “a third of the way in” and the field says 13.4.



Choice

**Popups, switches, checkboxes.**

Kind

Shape

Style

A choice can change which rows are shown: pick dot and Length disappears, because a dot has none.

Dial Rings Hands Date Export

Ring

Hours

24 hours

Hour marks

Minutes

numerals

numerals

markers

ticks

Add ring...

Duplicate

Delete

Up

Down

Kind

ticks

Colour

Outer radius

14.40

Positions

Count

60

Start °

0.0

Span °

360

Every

1

Offset

0

Skip those instead

Give way to rings above

Skip where

markers above

Knock out

nothing

Look

Shape

bar

wedge

dot

file

Length

1.00

Width · Ø

0.16

Round outer

0.00

Round inner

0.00

Minutes · Ø 30.0 mm · canvas · build 0 ms · frame 1 ms

Dial Tool · beta 2.3

5

10:09:36

Guides

Canvas

PDF

Now

Play

Fit

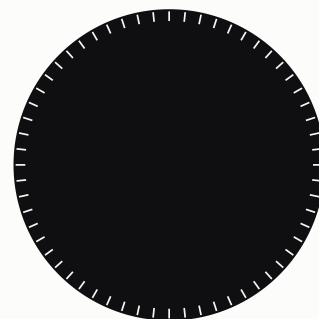
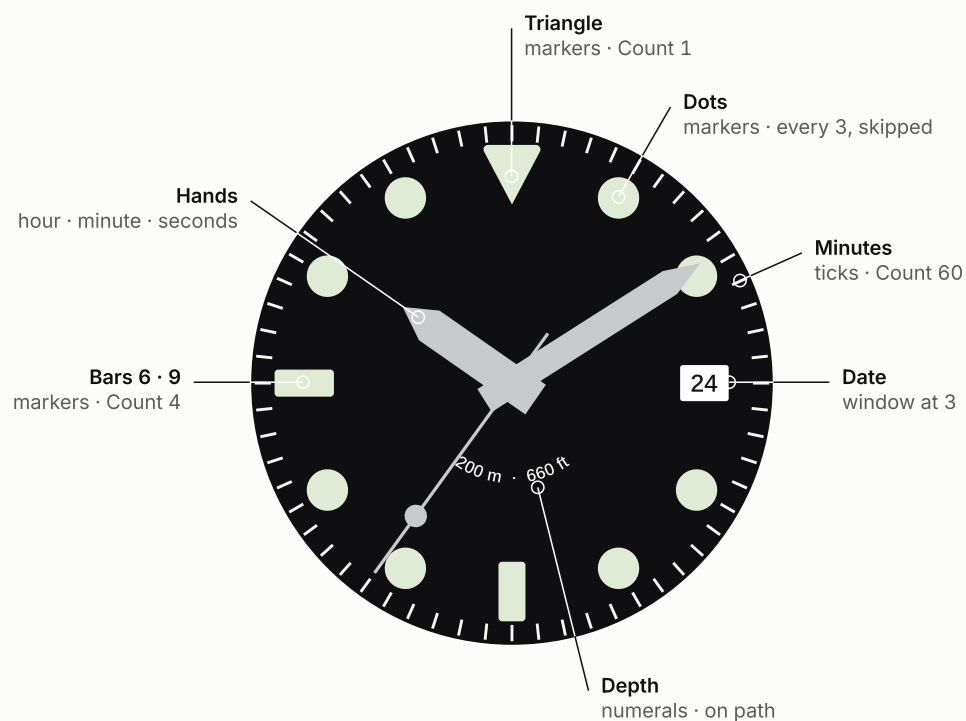
6

7

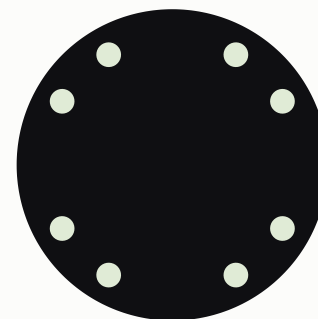
02

# Rings

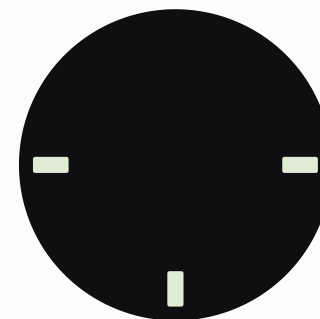
A ring repeats one thing round the centre, at its positions. A dial is a short list of rings, stacked.



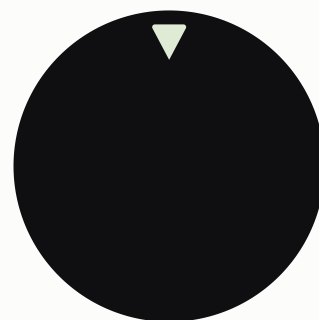
**Minutes**  
ticks · Count 60



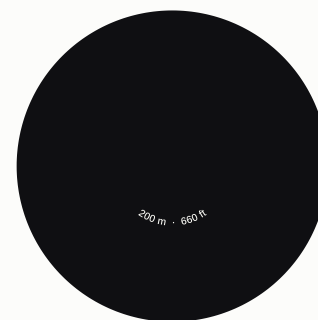
**Dots**  
markers · Every 3, skipped



**Bars 6 · 9**  
markers · Count 4 · 12: none



**Triangle**  
markers · Count 1 · wedge



**Depth**  
numerals · Custom · on path



**The dial**  
with hands and the date

## The diver's ring list

| Ring                                           | Kind     |
|------------------------------------------------|----------|
| <input checked="" type="checkbox"/> Triangle   | markers  |
| <input checked="" type="checkbox"/> Bars 6 · 9 | markers  |
| <input checked="" type="checkbox"/> Dots       | markers  |
| <input checked="" type="checkbox"/> Depth      | numerals |
| <input checked="" type="checkbox"/> Minutes    | ticks    |



### Watch

Order works twice. It is the stacking order, and a ring only gives way to rings **above** it. For ticks to stop at the markers, the markers go higher in the list.



### The idea

## One dial, five rings.

The diver of page 16, taken apart. Nothing is special-cased: the triangle is a markers ring with a count of one, the depth rating a numerals ring with one label.



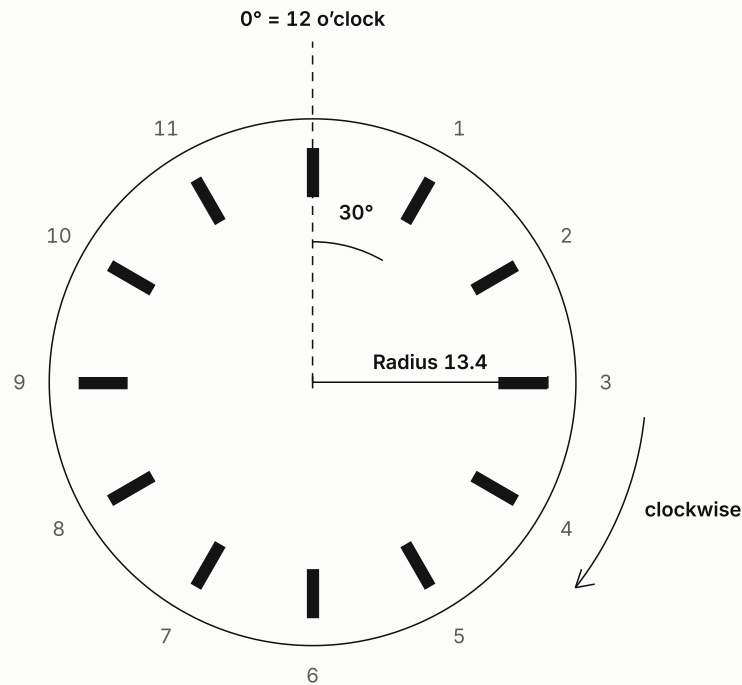
### Kind

## Four kinds.

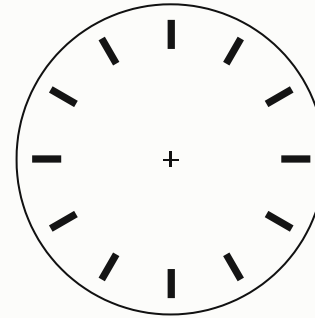
Ticks and markers repeat a shape, numerals a label; a band is colour between two radii. A ring can change kind at any time: it carries every setting with it.

# Positions

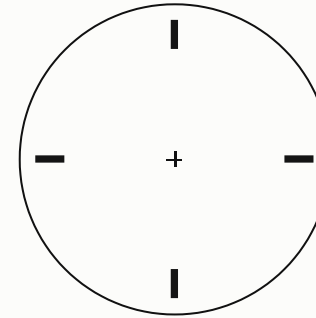
Where a ring puts things. Six numbers describe almost any track, scale or set of hours.



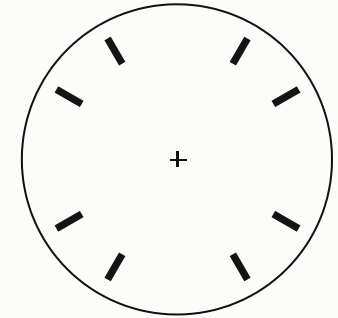
**Positions 0 – 11**  
Every and Offset count these



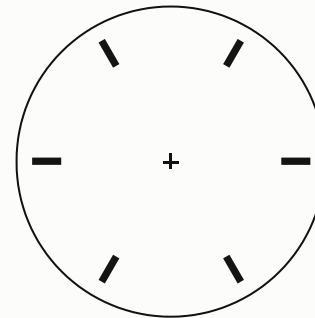
**Count 12**  
Start 0° · Span 360°



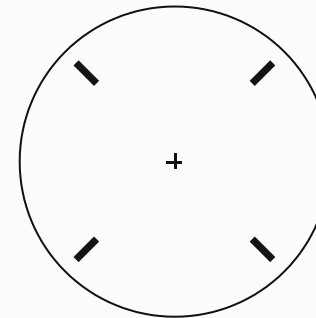
**Every 3**  
the cardinals



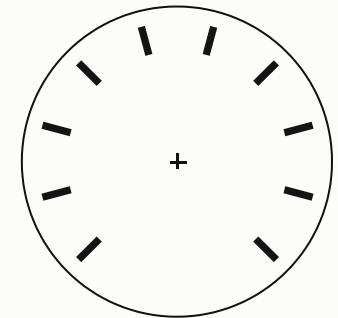
**Every 3 · skip those instead**  
the other eight



**Every 2 · Offset 1**  
the odd hours



**Count 4 · Start 45°**  
a quarter turn off



**Count 10 · Span 270°**  
Start -135° · both ends marked

## The arc, sixth picture

Count

Start °

Span °

## The cardinals, second picture

Every

Offset



## Watch

Radii are absolute. Changing the dial's Diameter moves no ring and no hand. Set the diameter first, then each Radius, the way a movement's drawing would.



Count · Start · Span

## An arc marks both ends.

Span 360° spaces the count evenly. Anything less puts the first and the last on the ends: Count 10 over 270° runs 0 to 9, as a gauge should.



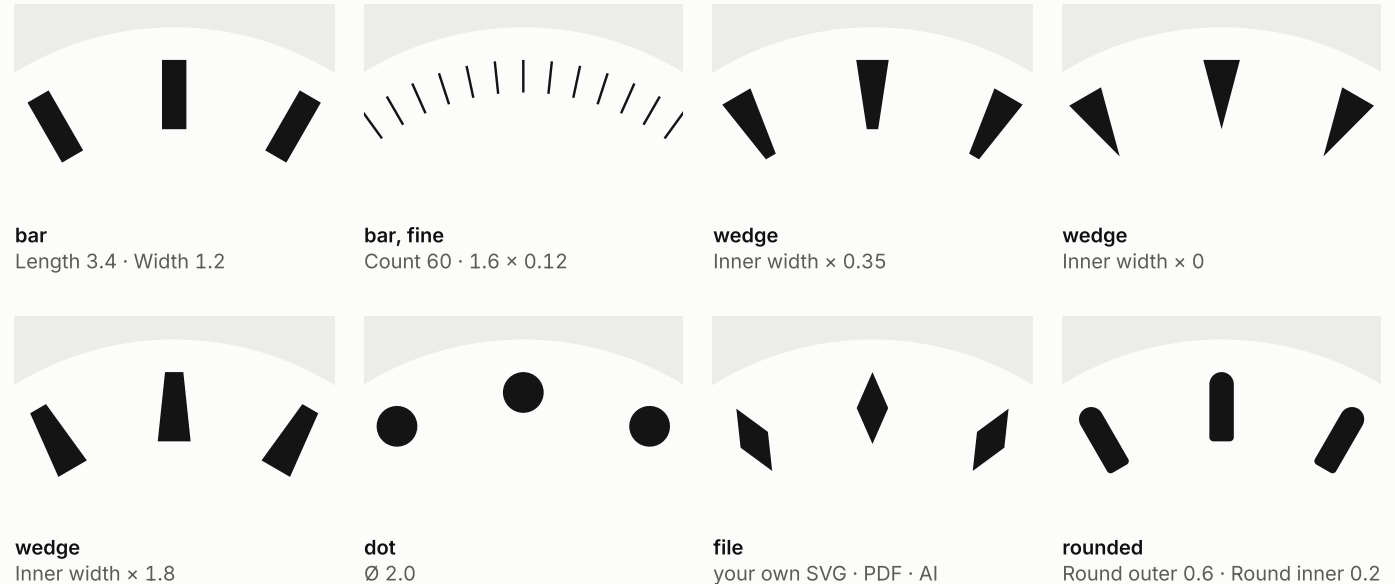
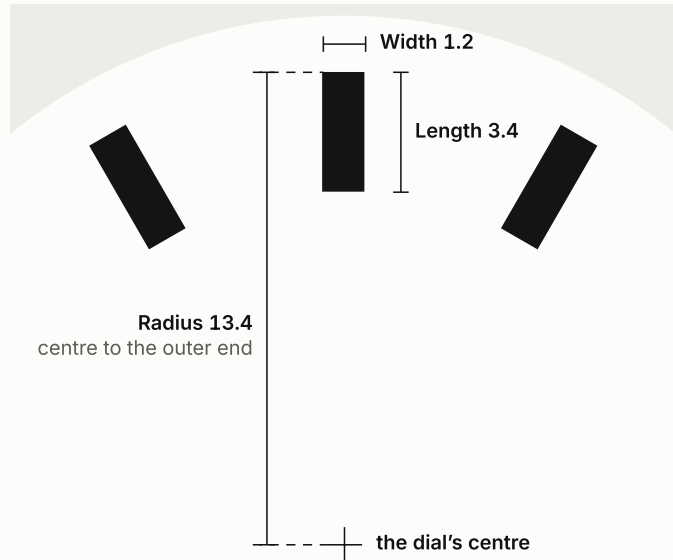
Every · Offset

## Use some of them.

Every 3 keeps the cardinals; Offset shifts which ones. Skip those instead keeps the other eight: two rings can share one circle without ever meeting.

# Marks

Ticks and markers share one set of shapes. A markers ring adds a first position that can differ from the rest.



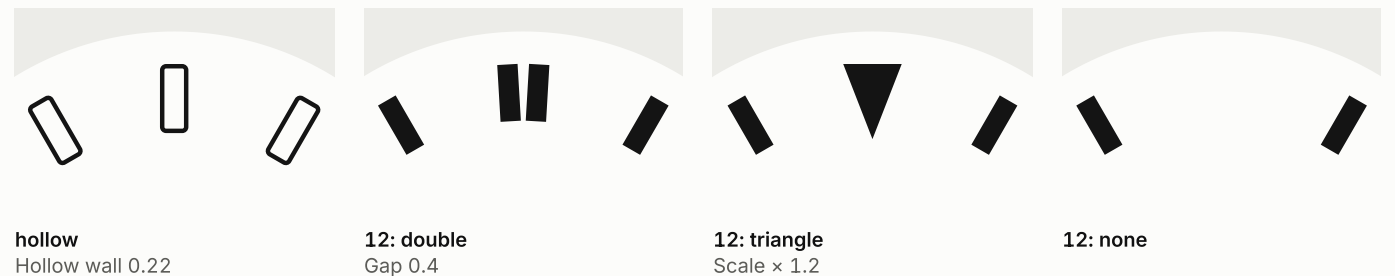
The mark above

Shape ☐ bar ☐ wedge ☐ dot ☐ file

Outer radius  13.40

Length  3.40

Width · Ø  1.20



## Watch

A fillet can't exceed half the side it rounds, and a Hollow wall of half the width or more leaves the mark solid. Neither warns: the shape just stops changing.



## Wedge and corners

### One number tapers; two round.

Inner width × is the inner end as a share of the outer: 0 a triangle, 1 a bar, more widens inward. Round outer and Round inner are true fillets, in mm.



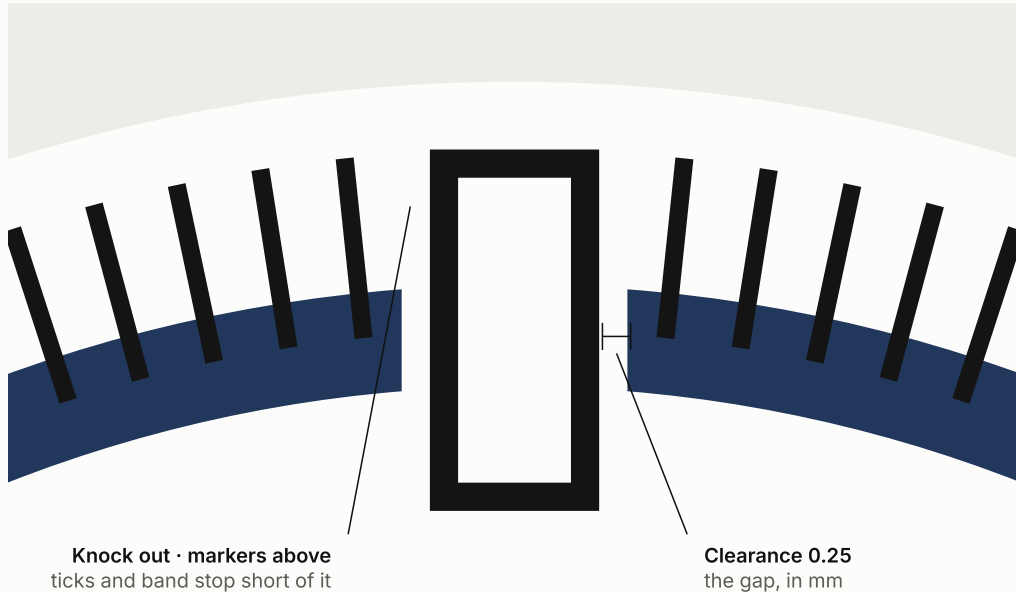
## 12 o'clock

### The first position can differ.

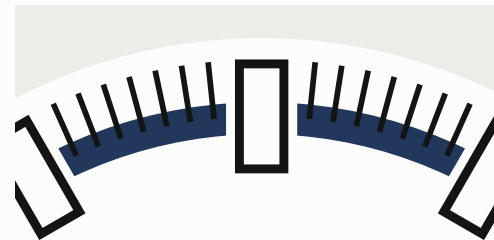
Markers draw their first position the same, double (with a Gap), as a triangle, or not at all. Scale × sizes it. Ticks have no such position.

# Give way

How a ring reacts to the rings above it. One mechanism covers every pair of rings.



skip where  
markers above



## The ticks above

Skip where

Knock out

Clearance



## Watch

Skip compares angles to a hundredth of a degree. Two rings skip each other only where their positions truly coincide: same Start, counts that divide. Otherwise use Knock out, which looks at shapes, not angles.



Skip where

## Leave out what is taken.

The ring drops its own positions wherever the chosen rings above already have a mark: minute ticks that stop for the hour markers.



Knock out

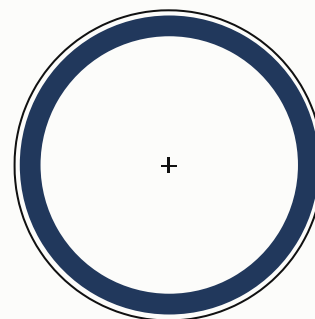
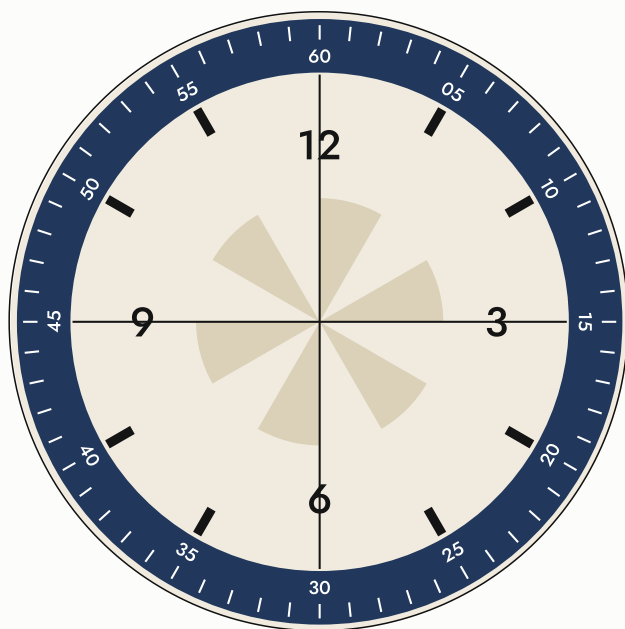
## Cut a gap round it.

The ring is cut back by Clearance mm round the shapes above: markers, markers and numerals, or everything. The cut follows the shape, serif and all.

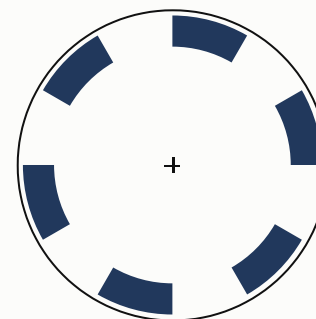


# Bands

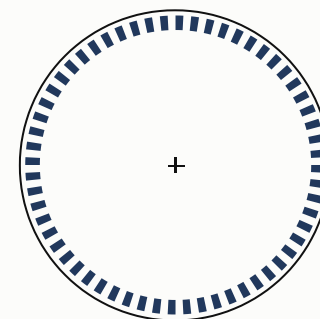
Flat colour between Radius and Radius – Band width: rings, rails, zones and sectors.



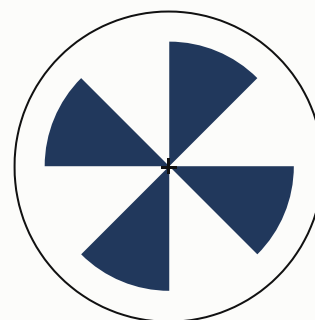
**ring**  
Count 1 · Band width 2.0



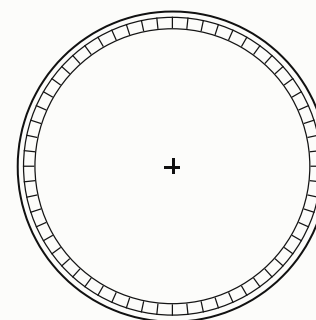
**sectors**  
Count 12 · Every 2



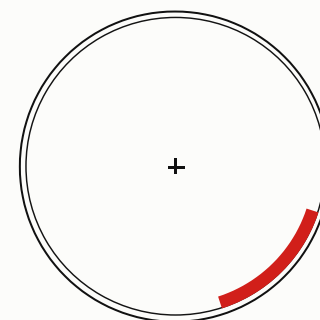
**fill**  
Count 60 · Fill 50 %



**pies**  
Band width = radius



**rails**  
two bands, Band width 0.12



**arc**  
Start 108° · Span 54°



Rails

**A hairline is a band too.**

Band width 0.12 draws a circle line. Two of them and a ticks ring make a railroad track: page 19.



Watch

Colour is flat, one per ring. A chapter ring with white print is three rings: the band, then white ticks and numerals above it, as in the sector dial here and on page 22.



Segments

**Positions cut a band.**

Count 1 is a full ring. More positions cut it into segments that tile the span; Fill % is how much of its step each one covers.

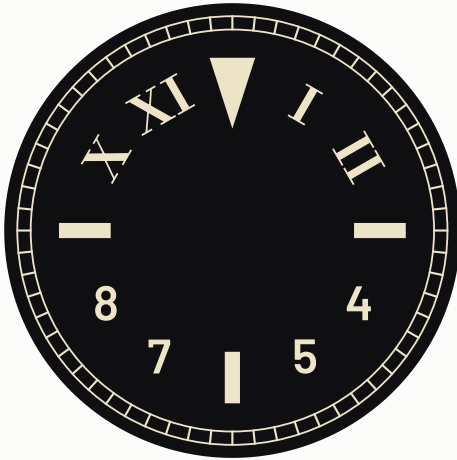


The sectors, second picture

|              |                                    |
|--------------|------------------------------------|
| Outer radius | <input type="text" value="14.40"/> |
| Band width   | <input type="text" value="3.00"/>  |
| Count        | <input type="text" value="12"/>    |
| Every        | <input type="text" value="2"/>     |

# Numerals

One label per position: from a built-in set, a count, or a list of your own.



11 12 1

Arabic

XI XII I

Roman · III

XI XII I

Roman · IV

١١ ١٢ ١

Eastern Arabic

+ - + = -

Chinese

ELEVEN TWELVE ONE

Words  
on path

55 60 05

Numbers  
First 5 · Step 5 · Digits 2

NNW N NNE

Custom  
N, NNE, , , E, ...

12

Numbers

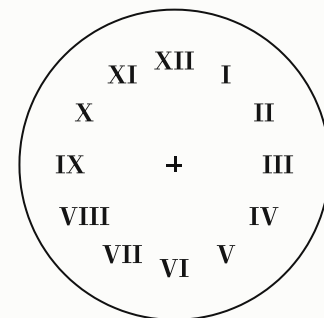
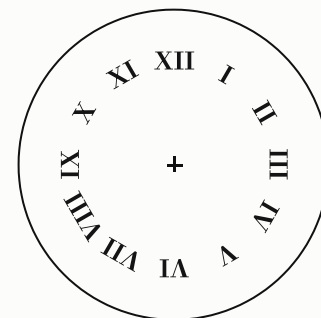
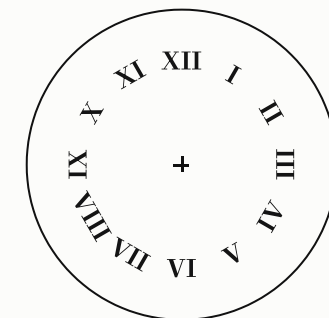
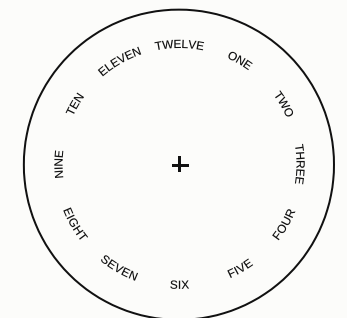
Labels that count.

Labels

First number

Step

Digits

upright  
always levelradial  
turns with the circleradial, auto-flip  
never upside downon path  
letters follow the circle

Watch

A Custom list follows positions, not hours. With Start 30° its first entry sits at 1 o'clock.



Placement

Four ways to sit.

On path sets each letter on the circle; below 3 and 9 it flips to read left to right.



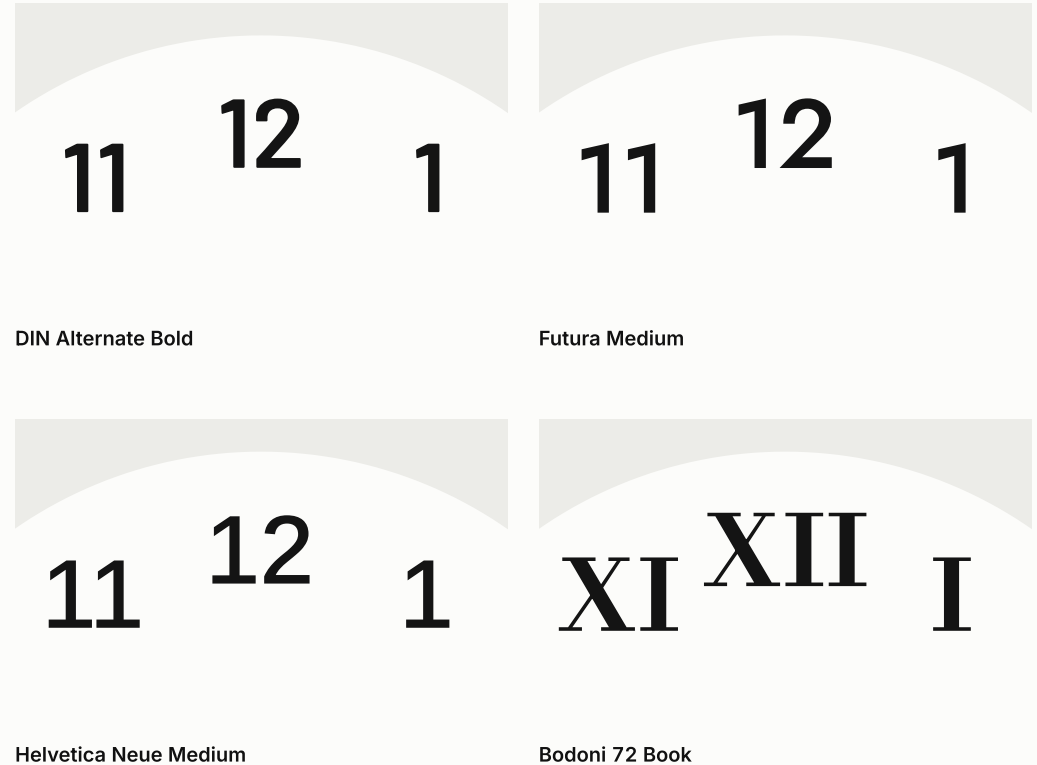
Custom

A list, comma by comma.

An empty entry leaves its position empty: California, page 19, is two such lists.

# Type and nudge

Each numerals ring has its own type, from any font on the Mac.  
Each numeral can then be corrected by eye.



Nudge

## One numeral at a time.

|          |                  |
|----------|------------------|
| Radius ± | in or out, mm    |
| Angle ±° | round the dial   |
| Rotate ° | about its centre |
| Size ×   | larger, smaller  |



Watch

Two-digit numerals look heavier than single ones. Size × 0.92 on 10, 11 and 12 evens them; Radius −0.2 pulls a wide 12 inward. Nudges belong to the position, not the label: change Count and they stay where they were.



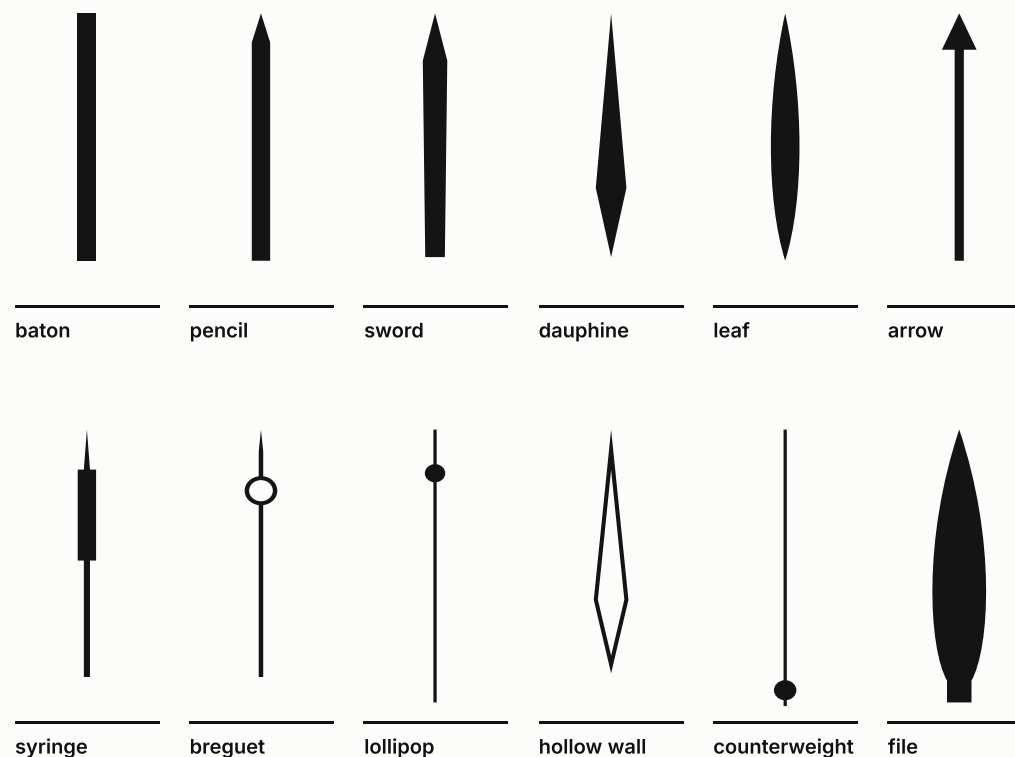
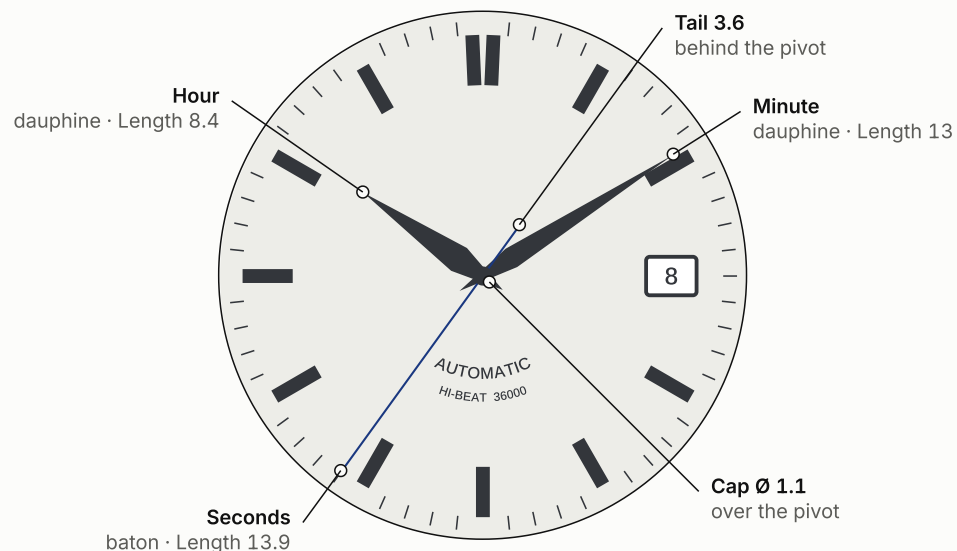
Type

## What reaches the outlines.

|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| Family · Style      | read from macOS; there is no fonts folder                       |
| Instance · Axes     | a variable font's named instances, and a slider per axis        |
| Features            | OpenType features, under the designer's own names               |
| Language · Tracking | local forms, such as Turkish İ i · 1/1000 em, as in Illustrator |

# Hands

Hour, minute and seconds each have their own shape, size and colour. Lengths are millimetres from the pivot.



Length · Tail

## Measured from the pivot.

Length runs from the pivot to the tip, Tail the other way. Counterweight Ø adds a disc to the tail; Hollow wall turns any hand into an outline.



Watch

Lengths don't follow the dial. A 13 mm minute hand on a 24 mm dial overhangs it by a millimetre, and nothing stops you. Neither does a real movement.



The diver's minute hand

|            |                       |       |
|------------|-----------------------|-------|
| Shape      | pencil                |       |
| Length     | <input type="range"/> | 12.40 |
| Width      | <input type="range"/> | 1.50  |
| Tail       | <input type="range"/> | 1.80  |
| Tip length | <input type="range"/> | 2.00  |



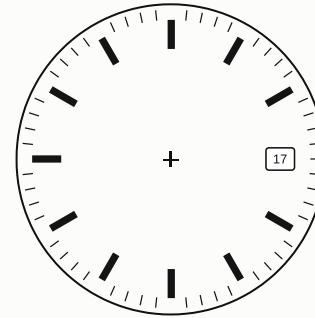
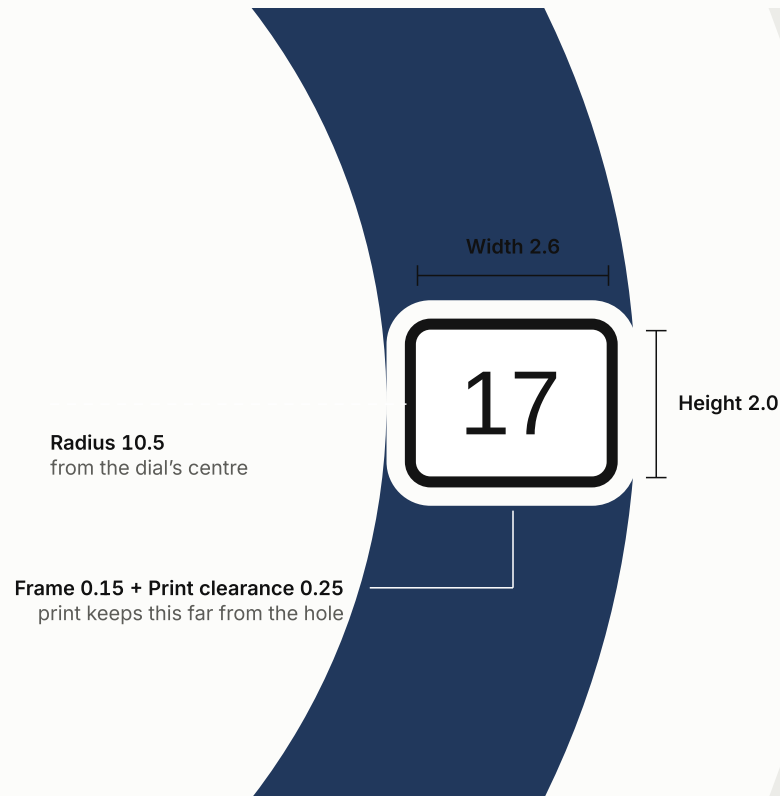
Motion

## Four beats.

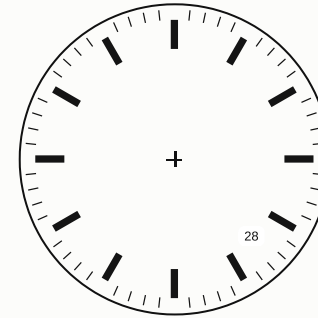
|               |                     |
|---------------|---------------------|
| sweep         | smooth              |
| quartz        | 1 step a second     |
| 6 · 8 beats/s | 21,600 · 28,800 vph |

# Date window

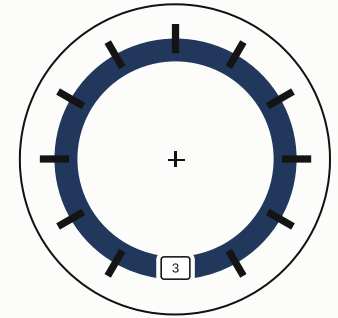
A hole in the plate that shows the date disc, with an optional printed frame. Print keeps its distance.



at 3  
2.6 × 2.0 · Frame 0.15



at 4:30  
Corner 0.6 · Frame 0



at 6  
Print clearance 0.4 cuts the band



Clearance

## Print keeps away.

Every ring is cut back by the frame plus Print clearance. Markers and numerals leave the date's position empty altogether.



Type

## The top numerals ring's font.

Shows is the day drawn on the disc; Text % of height sizes it. 28 is the widest day: check with it.

The window above

Position

Radius

Width

Height

Corner

Frame

Print clearance



Watch

The frame and the day share one Print colour, and the disc is always white. On a dark plate a light frame would mean a light day on white: set Frame to 0 there, as the diver does.



Production

## A cut line.

In a production mask the aperture is an outline to cut or drill to.



Position

## Five places.

3, 4:30, 6, 9 or 12; Radius sets how far out. The movement decides, not the layout.

# Classics

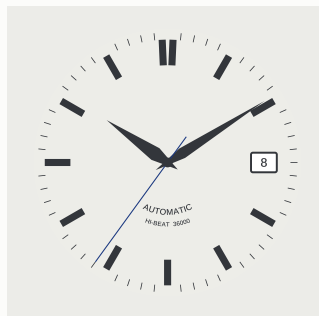
Nine dials after well-known types: approximations, not measured copies. Each page prints the settings that drew it.



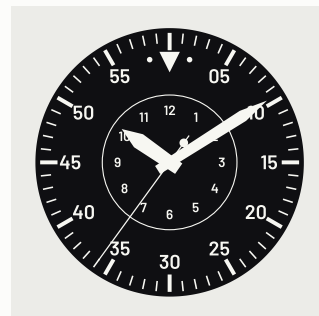
**Field watch**  
page 15



**Diver**  
page 16



**Dress watch**  
page 17



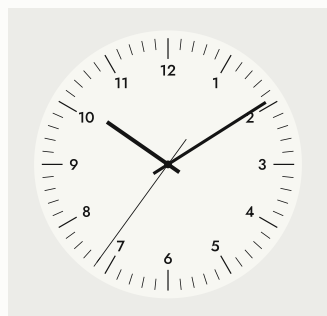
**Pilot, type B**  
page 18



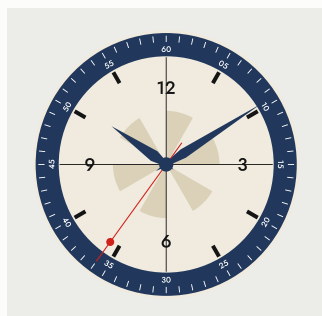
**California**  
page 19



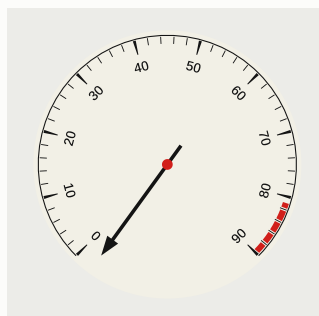
**Station clock**  
page 20



**Bauhaus**  
page 21



**Sector dial**  
page 22



**Gauge**  
page 23



Presets

## Load them.

Each dial is a settings file in *docs/guide/presets*. Export → Settings → Load... opens it, ready to change.



Reading a page

## Rings top first.

Only what differs from a newly added ring is printed; everything else is at its starting value.



How they were drawn

## By eye, not by caliper.

Proportions were judged against pictures of the type, not measured from a watch. Treat each as a starting point: load it, then move it toward yours.



Watch

Fonts are named as on macOS: DIN Alternate, Futura, Helvetica Neue, Bodoni 72. These pages were drawn with similar open fonts, so numerals differ slightly from what the presets draw on a Mac.



Watch

Load... replaces the whole dial: rings, hands and date. Save your own first, with Export → Settings → Save...



12.1

# Field watch

After the A-11 of the 1940s and its descendants: 12 and 24 hours, a plain minute track, nothing else.



Look for  
The 24-hour ring is a second numerals ring with a Custom list. The minute ticks **skip where** the hour marks sit, so the two never overprint.

Rings top first, then hands. Only what differs from a new ring.

|                    |                                                                                                                                                |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Hours numerals     | Radius 10.5 · Size 3 · Font DIN Alternate Bold <input type="radio"/>                                                                           |
| 24 hours numerals  | Radius 7.3 · Labels Custom · Custom 24, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23 · Size 1.35 · Font DIN Alternate Bold <input type="radio"/> |
| Hour marks markers | Radius 14.4 · Length 1.7 · Width 0.42 <input type="radio"/>                                                                                    |
| Minutes ticks      | Radius 14.4 · Count 60 · Length 1 · Width 0.16 · Skip where markers above <input type="radio"/>                                                |
| Hour hand          | Shape sword · Length 7.6 · Width 1.5 · Tail 1.6 · Tip length 2.4 <input type="radio"/>                                                         |
| Minute hand        | Shape sword · Length 12.6 · Width 1.2 · Tail 1.8 · Tip length 3 <input type="radio"/>                                                          |
| Seconds hand       | Shape baton · Length 13.2 · Width 0.14 · Tail 3.4 <input type="radio"/>                                                                        |
| Cap                | Ø 1.5 <input type="radio"/>                                                                                                                    |
| Preset             | presets/field.json · Dial 30 mm <input checked="" type="radio"/>                                                                               |



12.2

# Diver

After the 1950s dive watch: fat dots, bars at 6 and 9, a triangle at 12, the date at 3.



Look for  
Three markers rings share the hours: dots on **Every 3, skip those instead**; bars on Count 4 with the first position set to none; one triangle. The date takes 3 by itself.

Rings top first, then hands. Only what differs from a new ring.

|                    |                                                                                                                                        |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Triangle markers   | Radius 13.2 · Count 1 · Shape wedge · Length 3.3 · Width 3.5 · Inner width × 0 · Round outer 0.25                                      |
| Bars 6 · 9 markers | Radius 13.2 · Count 4 · Length 3.3 · Width 1.5 · Round outer 0.2 · Round inner 0.2 · 12 o'clock none                                   |
| Dots markers       | Radius 13 · Every 3 · Skip those instead · Shape dot · Ø 2.3                                                                           |
| Depth numerals     | Radius 5.2 · Count 1 · Start 180° · Labels Custom · Custom 200 m · 660 ft · Placement on path · Size 0.95 · Font Helvetica Neue Medium |
| Minutes ticks      | Radius 14.3 · Count 60 · Length 0.9 · Width 0.16                                                                                       |
| Date window        | Position 3 · Radius 10.7 · Width 2.7 · Height 2 · Corner 0.15 · Frame 0 · Print clearance 0.3 · Shows 24 · Text 68 %                   |
| Hour hand          | Shape pencil · Length 7.4 · Width 2 · Tail 1.6 · Tip length 1.8                                                                        |
| Minute hand        | Shape pencil · Length 12.4 · Width 1.5 · Tail 1.8 · Tip length 2                                                                       |
| Seconds hand       | Shape lollipop · Length 13 · Width 0.18 · Tail 3.4 · Head · ring · disc 1.25 · at 70 %                                                 |
| Cap                | Ø 1.6                                                                                                                                  |
| Preset             | presets/diver.json · Dial 29 mm                                                                                                        |





12.3

# Dress watch

After the Japanese dress watches of the 1960s: plain bars, a double bar at 12, dauphine hands, a framed date.



Look for  
The double bar is the markers ring's 12 o'clock style. Both lines of text are numerals rings with Count 1, set on path at 180°.

|                                                                 |                                                                                                                                      |
|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Rings top first, then hands. Only what differs from a new ring. |                                                                                                                                      |
| Hours markers                                                   | Radius 13.9 · Length 2.9 · Width 0.8 · 12 o'clock double · Gap 0.3 ●                                                                 |
| Minutes ticks                                                   | Radius 14.7 · Count 60 · Length 0.8 · Width 0.09 · Skip where markers above ●                                                        |
| Line 1 numerals                                                 | Radius 5.6 · Count 1 · Start 180° · Labels Custom · Custom AUTOMATIC · Placement on path · Size 0.95 · Font Helvetica Neue Medium ●  |
| Line 2 numerals                                                 | Radius 7 · Count 1 · Start 180° · Labels Custom · Custom HI-BEAT 36000 · Placement on path · Size 0.7 · Font Helvetica Neue Medium ● |
| Date window                                                     | Position 3 · Radius 10.9 · Width 2.7 · Height 2 · Corner 0.1 · Frame 0.22 · Print clearance 0.25 · Shows 8 · Text 70 %               |
| Hour hand                                                       | Shape dauphine · Length 8.4 · Width 1.6 · Tail 1.4 · at 20 % ●                                                                       |
| Minute hand                                                     | Shape dauphine · Length 13 · Width 1.25 · Tail 1.6 · at 15 % ●                                                                       |
| Seconds hand                                                    | Shape baton · Length 13.9 · Width 0.12 · Tail 3.6 ●                                                                                  |
| Cap                                                             | Ø 1.1 ●                                                                                                                              |
| Preset                                                          | presets/dress.json · Dial 30.4 mm ○                                                                                                  |



12.4

## Pilot, type B

After the 1940s observer's watch: minutes outside, hours on an inner ring, a triangle and two dots at 12.



Look for

The minutes are **Numbers**: first 5, step 5, two digits, over a 300° span that leaves 12 free for the triangle. The two dots are a ticks ring with Count 2 on a 22° span.

Rings top first, then hands. Only what differs from a new ring.

|                                |                                                                                                                                                                                                                   |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Minutes</b><br>numerals     | Radius <b>12.6</b> · Count <b>11</b> · Start <b>30°</b> · Span <b>300°</b> · Labels <b>Numbers</b> · First number <b>5</b> · Step <b>5</b> · Digits <b>2</b> · Size <b>2.5</b> · Font <b>DIN Alternate Bold</b> ○ |
| <b>Triangle</b><br>markers     | Radius <b>14</b> · Count <b>1</b> · Shape <b>wedge</b> · Length <b>2.6</b> · Width <b>2.6</b> · Inner width × <b>0</b> ○                                                                                          |
| <b>Two dots</b><br>ticks       | Radius <b>13.6</b> · Count <b>2</b> · Start <b>-11°</b> · Span <b>22°</b> · Shape <b>dot</b> · Ø <b>0.7</b> ○                                                                                                     |
| <b>Five minutes</b><br>markers | Radius <b>16.4</b> · Length <b>2.2</b> · Width <b>0.5</b> ○                                                                                                                                                       |
| <b>Minute track</b><br>ticks   | Radius <b>16.4</b> · Count <b>60</b> · Length <b>1.3</b> · Width <b>0.2</b> · Skip where <b>markers above</b> ○                                                                                                   |
| <b>Hours</b><br>numerals       | Radius <b>6.6</b> · Size <b>1.7</b> · Font <b>DIN Alternate Bold</b> ○                                                                                                                                            |
| <b>Hour circle</b><br>band     | Radius <b>8.6</b> · Band width <b>0.14</b> ○                                                                                                                                                                      |
| <b>Hour</b><br>hand            | Shape <b>sword</b> · Length <b>7.4</b> · Width <b>2</b> · Tail <b>1.4</b> · Tip length <b>2.6</b> ○                                                                                                               |
| <b>Minute</b><br>hand          | Shape <b>sword</b> · Length <b>14.6</b> · Width <b>1.5</b> · Tail <b>1.8</b> · Tip length <b>3.6</b> ○                                                                                                            |
| <b>Seconds</b><br>hand         | Shape <b>baton</b> · Length <b>15.4</b> · Width <b>0.16</b> · Tail <b>4.2</b> · Counterweight Ø <b>1.1</b> ○                                                                                                      |
| <b>Cap</b>                     | Ø <b>1.7</b> ○                                                                                                                                                                                                    |
| <b>Preset</b>                  | <i>presets/pilot.json</i> · Dial <b>34 mm</b> ●                                                                                                                                                                   |



12.5

# California

After the 1930s–40s dial: Roman on top, Arabic below, bars at 3 · 6 · 9, a triangle at 12, a railroad track.



**Look for**  
Two numerals rings on one radius, each with a **Custom** list that leaves the other's positions empty — so Roman and Arabic can use different fonts and placements.

Rings top first, then hands. Only what differs from a new ring.

|                                |                                                                                                                                                                                 |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Roman numerals</b>          | Radius <b>9.6</b> · Labels <b>Custom</b> · Custom , I, II, , , , , , X, XI · Placement <b>radial</b> · Size <b>3.3</b> · Font <b>Bodoni 72 Book</b> <input type="radio"/>       |
| <b>Arabic numerals</b>         | Radius <b>9.6</b> · Labels <b>Custom</b> · Custom , , , , , 4, 5, , 7, 8, , · Placement <b>upright</b> · Size <b>3.2</b> · Font <b>DIN Alternate Bold</b> <input type="radio"/> |
| <b>Bars + triangle markers</b> | Radius <b>11.4</b> · Count <b>4</b> · Length <b>3.4</b> · Width <b>1</b> · 12 o'clock <b>triangle</b> · Scale × <b>1.25</b> <input type="radio"/>                               |
| <b>Minutes ticks</b>           | Radius <b>14.2</b> · Count <b>60</b> · Length <b>1</b> · Width <b>0.14</b> <input type="radio"/>                                                                                |
| <b>Outer rail band</b>         | Radius <b>14.2</b> · Band width <b>0.12</b> <input type="radio"/>                                                                                                               |
| <b>Inner rail band</b>         | Radius <b>13.32</b> · Band width <b>0.12</b> <input type="radio"/>                                                                                                              |
| <b>Hour hand</b>               | Shape <b>pencil</b> · Length <b>7.2</b> · Width <b>1.5</b> · Tail <b>1.4</b> · Tip length <b>1.6</b> <input type="radio"/>                                                      |
| <b>Minute hand</b>             | Shape <b>pencil</b> · Length <b>11.8</b> · Width <b>1.2</b> · Tail <b>1.6</b> · Tip length <b>1.8</b> <input type="radio"/>                                                     |
| <b>Seconds hand</b>            | Shape <b>baton</b> · Length <b>12.6</b> · Width <b>0.13</b> · Tail <b>3.2</b> · Counterweight Ø <b>0.9</b> <input type="radio"/>                                                |
| <b>Cap</b>                     | Ø <b>1.4</b> <input type="radio"/>                                                                                                                                              |
| <b>Preset</b>                  | <i>presets/california.json</i> · Dial <b>30 mm</b> <input checked="" type="radio"/>                                                                                             |



12.6

# Station clock

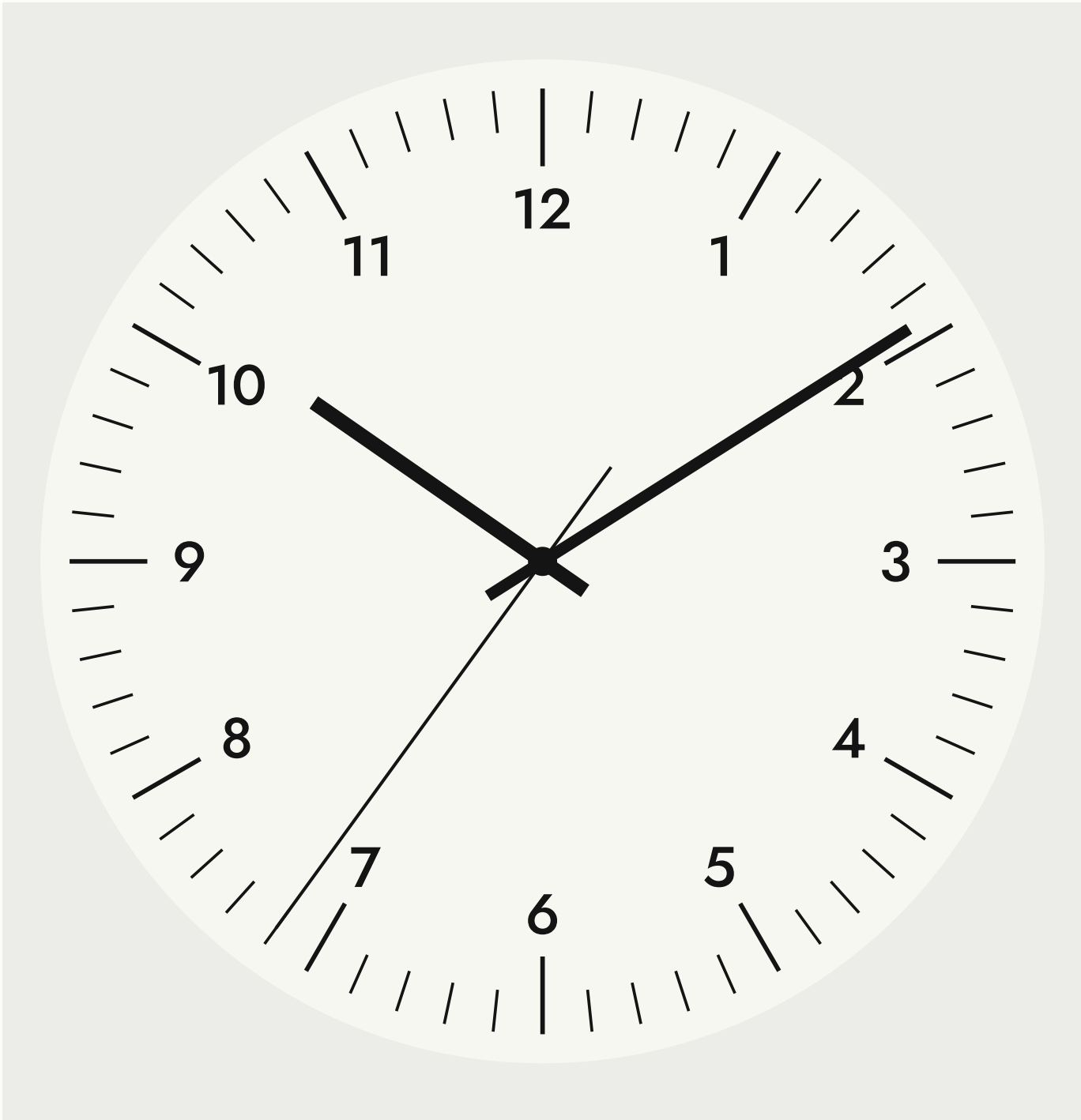
After the Swiss railway clock of 1944:  
heavy bars, and a red seconds hand like a  
dispatcher's baton.



Look for  
Two rings. Everything is in the proportions: bars  $3.7 \times 1.3$ ,  
ticks  $1.3 \times 0.45$ , and a lollipop with its disc at 100 % of the  
length.

Rings top first, then hands. Only what differs from a new ring.

|                  |                                                                                                                                        |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Hours<br>markers | Radius <b>14.3</b> · Length <b>3.7</b> · Width <b>1.3</b> ●                                                                            |
| Minutes<br>ticks | Radius <b>14.3</b> · Count <b>60</b> · Length <b>1.3</b> · Width <b>0.45</b> ·<br>Skip where <b>markers above</b> ●                    |
| Hour<br>hand     | Shape <b>baton</b> · Length <b>8.8</b> · Width <b>1.7</b> · Tail <b>2.4</b> ●                                                          |
| Minute<br>hand   | Shape <b>baton</b> · Length <b>13.6</b> · Width <b>1.3</b> · Tail <b>2.8</b> ●                                                         |
| Seconds<br>hand  | Shape <b>lollipop</b> · Length <b>9.8</b> · Width <b>0.34</b> · Tail <b>3.6</b> ·<br>Head · ring · disc <b>2.7</b> · at <b>100 %</b> ● |
| Cap              | Ø <b>1</b> ●                                                                                                                           |
| Preset           | <i>presets/railway.json</i> · Dial 30 mm ○                                                                                             |



12.7

# Bauhaus

After the German wall-clock school of the 1950s–60s: hairline marks, small numerals, thin hands.



Look for  
Line weights carry it: 0.14 for the hours, 0.09 for the minutes, hands under half a millimetre.

|                                                                 |                                                                               |
|-----------------------------------------------------------------|-------------------------------------------------------------------------------|
| Rings top first, then hands. Only what differs from a new ring. |                                                                               |
| Hours numerals                                                  | Radius 10.9 · Size 1.75 · Font Futura Medium ●                                |
| Hour lines markers                                              | Radius 14.6 · Length 2.4 · Width 0.14 ●                                       |
| Minutes ticks                                                   | Radius 14.6 · Count 60 · Length 1.3 · Width 0.09 · Skip where markers above ● |
| Hour hand                                                       | Shape baton · Length 8.6 · Width 0.46 · Tail 1.6 ●                            |
| Minute hand                                                     | Shape baton · Length 13.4 · Width 0.36 · Tail 2 ●                             |
| Seconds hand                                                    | Shape baton · Length 14 · Width 0.1 · Tail 3.6 ●                              |
| Cap                                                             | Ø 0.9 ●                                                                       |
| Preset                                                          | presets/bauhaus.json · Dial 31 mm ○                                           |

# Sector dial

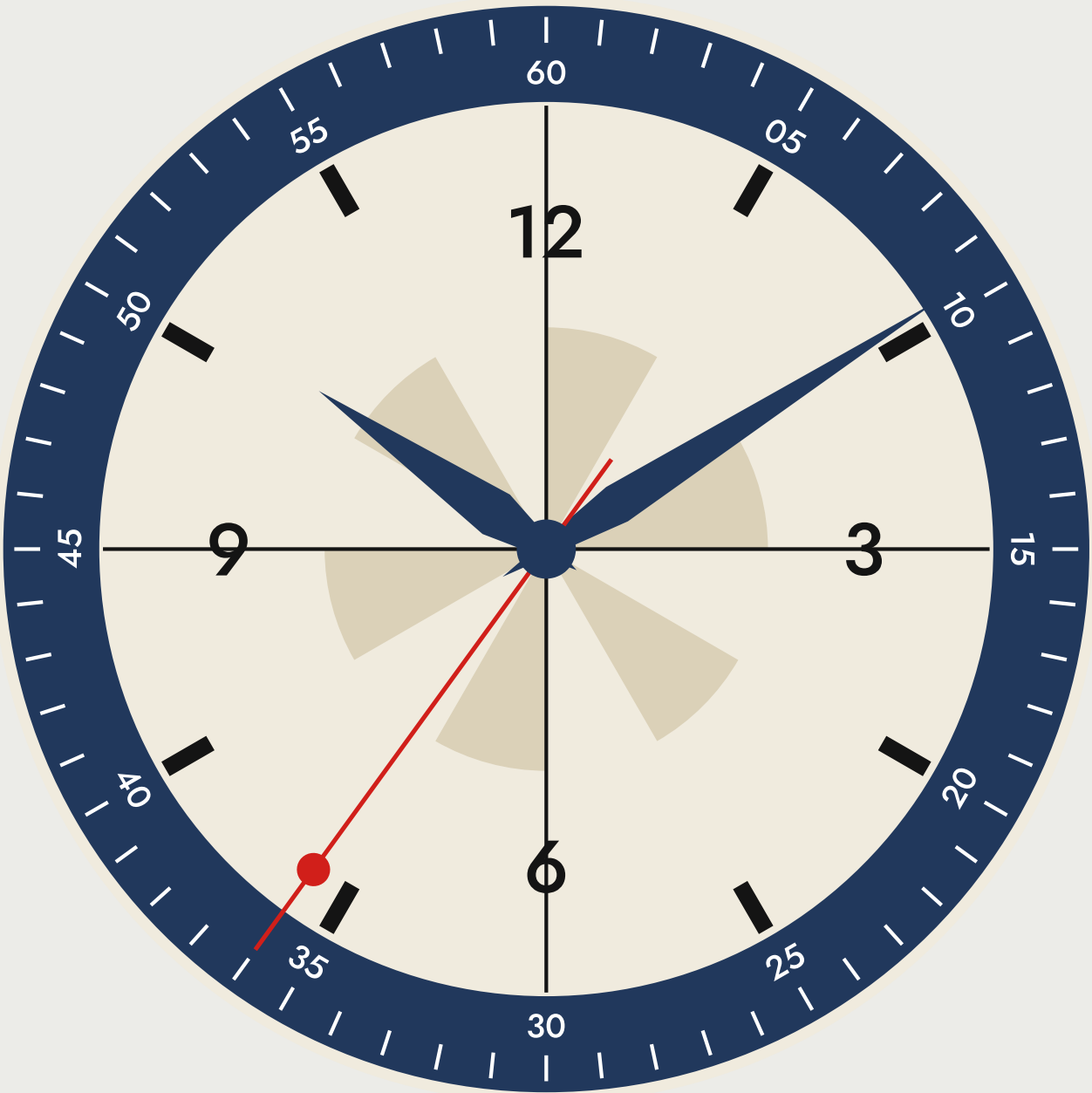
After the 1930s scientific dials: a chapter ring, a crosshair, zones of colour.



Look for  
Bands do the colour: a chapter ring 2.6 wide, and pies made by Band width = Radius with Every 2. The crosshair is a ticks ring, Count 4, Length 12.

Rings top first, then hands. Only what differs from a new ring.

|                          |                                                                                                                                               |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Hours numerals           | Radius 8.6 · Every 3 · Size 2 · Font Futura Medium ●                                                                                          |
| Hour bars markers        | Radius 11.9 · Every 3 · Skip those instead · Length 1.4 · Width 0.45 ●                                                                        |
| Minutes ticks            | Radius 14.4 · Count 60 · Length 0.7 · Width 0.1 ○                                                                                             |
| Minute numerals numerals | Radius 12.9 · Start 30° · Labels Numbers · First number 5 · Step 5 · Digits 2 · Placement radial, auto-flip · Size 0.9 · Font Futura Medium ○ |
| Chapter band             | Radius 14.7 · Band width 2.6 ●                                                                                                                |
| Crosshair ticks          | Radius 12 · Count 4 · Length 12 · Width 0.1 ●                                                                                                 |
| Sectors band             | Radius 6 · Count 12 · Every 2 · Band width 6 ○                                                                                                |
| Hour hand                | Shape dauphine · Length 7.5 · Width 1.3 · Tail 1 · at 22 % ●                                                                                  |
| Minute hand              | Shape dauphine · Length 12.6 · Width 1.1 · Tail 1.4 · at 18 % ●                                                                               |
| Seconds hand             | Shape lollipop · Length 13.4 · Width 0.12 · Tail 3 · Head · ring · disc 0.9 · at 80 % ●                                                       |
| Cap                      | Ø 1.6 ●                                                                                                                                       |
| Preset                   | presets/sector.json · Dial 30 mm ○                                                                                                            |





12.9

# Gauge

A 270° scale with a red zone: positions over a partial span mark both ends.



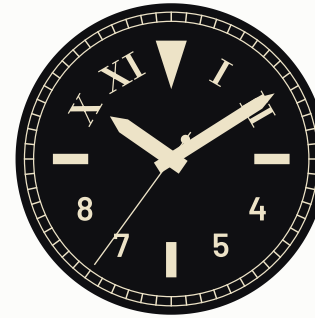
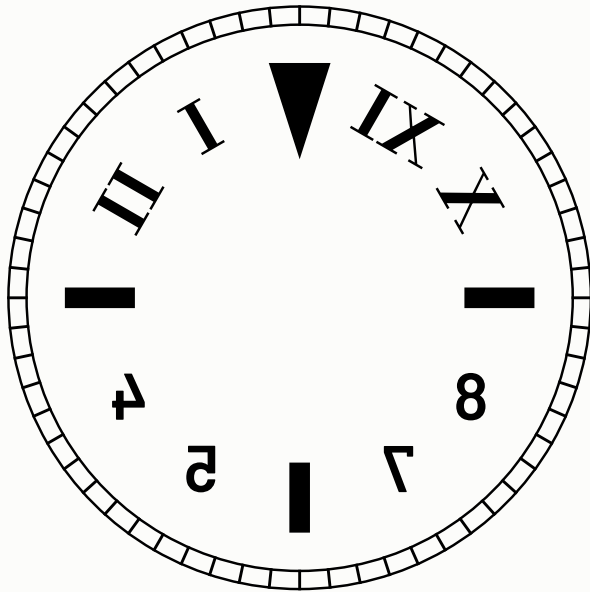
Look for  
Start -135°, Span 270°. A partial span marks both ends, so Count 10 gives 0 to 90. The red zone **knocks out** everything above it with a 0.12 clearance.

Rings top first, then hands. Only what differs from a new ring.

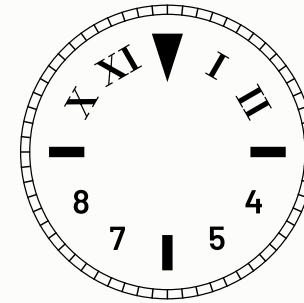
|                     |                                                                                                                                                                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scale numerals      | Radius <b>11.3</b> · Count <b>10</b> · Start <b>-135°</b> · Span <b>270°</b> · Labels <b>Numbers</b> · First number <b>0</b> · Step <b>10</b> · Placement <b>radial, auto-flip</b> · Size <b>1.5</b> · Font <b>Helvetica Neue Medium</b> ● |
| Scale marks markers | Radius <b>14.3</b> · Count <b>10</b> · Start <b>-135°</b> · Span <b>270°</b> · Shape <b>wedge</b> · Length <b>1.6</b> · Width <b>0.34</b> · Inner width × <b>0.3</b> ●                                                                     |
| Scale ticks ticks   | Radius <b>14.3</b> · Count <b>46</b> · Start <b>-135°</b> · Span <b>270°</b> · Length <b>0.8</b> · Width <b>0.1</b> · Skip where <b>markers above</b> ●                                                                                    |
| Rail band           | Radius <b>14.5</b> · Count <b>1</b> · Start <b>-135°</b> · Span <b>270°</b> · Band width <b>0.12</b> ●                                                                                                                                     |
| Red zone band       | Radius <b>14.5</b> · Count <b>1</b> · Start <b>108°</b> · Span <b>27°</b> · Band width <b>0.9</b> · Knock out <b>everything above</b> · Clearance <b>0.12</b> ●                                                                            |
| Minute hand         | Shape <b>arrow</b> · Length <b>12.6</b> · Width <b>0.4</b> · Tail <b>2.6</b> · Tip length <b>2.2</b> · Head · ring · disc <b>1.5</b> ●                                                                                                     |
| Cap                 | Ø <b>1.2</b> ●                                                                                                                                                                                                                             |
| Preset              | <i>presets/gauge.json</i> · Dial <b>30 mm</b> ○                                                                                                                                                                                            |

# Files in and out

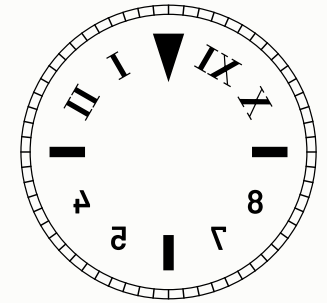
Vectors at true size, a one-colour mask for production, and your own shapes coming in.



**design**  
PDF · SVG · PNG



**production**  
one colour, no plate, no hands



**production + mirror**  
for toner transfer

## Export

☒ Production: mono print mask

☒ Mirror: toner transfer

☐ Live text in PDF · SVG (fonts n...)

PNG dpi

Clip seconds

Frames per s



PDF · SVG · PNG · MP4 · GIF

## True size.

A 30 mm dial opens 30 mm wide. PNG at your dpi: 30 mm at 600 dpi is 709 pixels. MP4 and GIF run the hands for Clip seconds.



Production · Mirror

## One colour, flipped.

Production drops the plate and the hands and prints every ring in one colour: a mask. Mirror flips it for toner transfer, where the print lands face down.



## Watch

This is that mask, mirrored. Check its thinnest line at 100 % before you etch: a 0.09 mm tick is two pixels at 600 dpi, and brass is less forgiving than a screen.



Live text

## Off by default.

Keeps numerals editable. The fonts are then needed where the file is opened, and on-path text becomes one object per letter.



In: SVG · PDF · AI

## Your own shapes.

Draw pointing up; the artboard's centre is the anchor. Black is ink, white cuts holes. Outline text first.



# Reference

Constructions to build on, every setting with its range, and a way in when you know what you want but not its name.



Track

## Railroad

Ticks, Count 60. Two bands 0.12 wide: at the ticks' Radius, and at Radius - Length + 0.12. Page 19.



Track

## Fifths of a second

Ticks Count 60 above ticks Count 300. The 300 Skip where: all marks above.



Hours

## 24-hour ring

Numerals, Labels Custom: 24, 13, 14 ... 23, at a smaller Radius and Size. Page 15.



Hours

## Two fonts, one circle

Two numerals rings on one Radius; each Custom list leaves the other's positions empty. Page 19.



Minutes

## 05 to 60

Numerals, Labels Numbers, First number 5, Step 5, Digits 2, Start 30°. Page 22.



Text

## A wordmark

Numerals, Count 1, Start 0° or 180°, Labels Custom, Placement on path. Page 17.



Colour

## A gap round numerals

A band below the numerals; Knock out: markers & numerals above; Clearance 0.3.



Colour

## Sector dial

A band, Count 12, Every 2, Band width equal to Radius. Page 22.



When something is wrong

## Read the log, compare the previews, send both.

- 1 Open Export and read the log. At launch it says "preview: canvas" and "all controls connected".
- 2 If the preview looks wrong, switch Canvas to PDF and compare.
- 3 Send a screenshot and the log. If the window doesn't build, run `tools/dial_probe.py` and send its report.



A ring is gone

Look at its checkbox in the list, its Every and Offset, and whether it skips positions a ring above takes.



No numeral at 3

The date window took it: markers and numerals leave its position empty.



Where things live

## One repository.

|               |                                                                                 |
|---------------|---------------------------------------------------------------------------------|
| Code and docs | <a href="https://github.com/duztman/dial-tool">github.com/duztman/dial-tool</a> |
| Words · rules | <a href="#">docs/glossary.md</a> · <a href="#">docs/design.md</a>               |
| Presets       | <a href="#">docs/guide/presets</a>                                              |
| This book     | <a href="#">docs/guide/pages</a> , built by <code>tools/make_guide.py</code>    |



Slow

The log should say "shape combining: skia-pathops". If not, install it: page 2.



Start over

Export → Settings → Defaults. The last session is kept in `~/Library/Application Support/DialTool`.

# Every ring setting

Printed from the tool's own rows, so it cannot drift. Starts at: a markers ring added to a 30 mm dial.

| A ring                  | From · to                                                                    | Step | Starts at | Shown for                           |
|-------------------------|------------------------------------------------------------------------------|------|-----------|-------------------------------------|
| Kind                    | ticks · <b>markers</b> · numerals · band                                     |      |           | all                                 |
| Colour                  | a colour                                                                     |      | ●         | all                                 |
| Outer radius            | 0 to 25                                                                      | 0.05 | 13.40     | ticks · markers · band              |
| Centre radius           | 0 to 25                                                                      | 0.05 | 13.40     | numerals                            |
| Positions               |                                                                              |      |           |                                     |
| Count                   | 1 to 360                                                                     | 1    | 12        | all                                 |
| Start °                 | -180 to 180                                                                  | 0.5  | 0.0       | all                                 |
| Span °                  | 1 to 360                                                                     | 1    | 360       | all                                 |
| Every                   | 1 to 60                                                                      | 1    | 1         | all                                 |
| Offset                  | 0 to 59                                                                      | 1    | 0         | all                                 |
| Skip those instead      | on · off                                                                     |      | off       | all                                 |
| Give way to rings above |                                                                              |      |           |                                     |
| Skip where              | <b>nothing</b> · markers above · markers & numerals above · all marks above  |      |           | all                                 |
| Knock out               | <b>nothing</b> · markers above · markers & numerals above · everything above |      |           | all                                 |
| Clearance               | 0 to 2                                                                       | 0.01 | 0.25      | all                                 |
| Look                    |                                                                              |      |           |                                     |
| Shape                   | bar · wedge · dot · file                                                     |      |           | ticks · markers                     |
| File                    | SVG · PDF · AI                                                               |      |           | ticks · markers (file)              |
| Size x                  | 0.05 to 10                                                                   | 0.01 | 1.00      | ticks · markers (file)              |
| Length                  | 0.05 to 10                                                                   | 0.05 | 2.80      | ticks · markers (bar · wedge)       |
| Width · Ø               | 0.02 to 5                                                                    | 0.01 | 0.70      | ticks · markers (bar · wedge · dot) |
| Inner width x           | 0 to 2                                                                       | 0.01 | 0.35      | ticks · markers (wedge)             |
| Round outer             | 0 to 2                                                                       | 0.01 | 0.00      | ticks · markers (bar · wedge)       |
| Round inner             | 0 to 2                                                                       | 0.01 | 0.00      | ticks · markers (bar · wedge)       |
| Hollow wall             | 0 to 1.5                                                                     | 0.01 | 0.00      | ticks · markers                     |

| A ring · 2                  | From · to                                                                               | Step | Starts at | Shown for |
|-----------------------------|-----------------------------------------------------------------------------------------|------|-----------|-----------|
| 12 o'clock · first position |                                                                                         |      |           |           |
| Style                       | same · double · triangle · none                                                         |      |           | markers   |
| Scale x                     | 0.5 to 2.5                                                                              | 0.05 | 1.00      | markers   |
| Gap                         | 0 to 3                                                                                  | 0.05 | 0.50      | markers   |
| Labels                      | Arabic · Roman · III · Roman · IV · Eastern Arabic · Chinese · Words · Custom · Numbers |      |           | numerals  |
| Custom                      | text                                                                                    |      |           | numerals  |
| First number                | −1000 to 1000                                                                           | 1    | 0         | numerals  |
| Step                        | −100 to 100                                                                             | 1    | 5         | numerals  |
| Digits                      | 0 to 4                                                                                  | 1    | 0         | numerals  |
| Placement                   | upright · radial · radial, auto-flip · on path                                          |      |           | numerals  |
| Size                        | 0.3 to 8                                                                                | 0.05 | 2.40      | numerals  |
| Band width                  | 0.02 to 25                                                                              | 0.05 | 1.00      | band      |
| Fill % of step              | 1 to 100                                                                                | 1    | 100       | band      |
| Nudge one numeral           |                                                                                         |      |           |           |
| Nudge                       | per numeral: Radius ± · Angle ±° · Rotate ° · Size x                                    |      |           | numerals  |
| Type                        |                                                                                         |      |           |           |
| Family                      | every font installed on the Mac                                                         |      |           | numerals  |
| Style                       | the family's styles                                                                     |      |           | numerals  |
| Instance                    | a variable font's named instances                                                       |      |           | numerals  |
| Language                    | default · tr · nl · de · pl · ro · ca · ar · fa · zh-Hans · ja                          |      |           | numerals  |
| Tracking                    | −200 to 800                                                                             | 5    | 0         | numerals  |
| Features                    | the font's OpenType features                                                            |      |           | numerals  |
| Axes                        | one slider per variable axis                                                            |      |           | numerals  |

# Every other setting

Dial, hands, date and export. A slider's range is where it stops, not where the value must: type past it.

| Hands              | From · to                                                                              | Step | Starts at                                                                           | Shown for                                  |
|--------------------|----------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------|--------------------------------------------|
| Show hands         | on · off                                                                               |      | on                                                                                  |                                            |
| Motion             | sweep · quartz · 1/s · 6 beats/s · 8 beats/s                                           |      |                                                                                     |                                            |
| Cap Ø              | 0 to 4                                                                                 | 0.05 | 1.60                                                                                |                                            |
| Cap colour         | a colour                                                                               |      |    |                                            |
| Hand               |                                                                                        |      |                                                                                     |                                            |
| Show this hand     | on · off                                                                               |      | on                                                                                  | all                                        |
| Shape              | baton · pencil · sword · dauphine · leaf · arrow · syringe · breguet · lollipop · file |      |                                                                                     | all                                        |
| File               | SVG · PDF · AI                                                                         |      |                                                                                     | file                                       |
| Fit file to length | on · off                                                                               |      | on                                                                                  | file                                       |
| Size x             | 0.05 to 10                                                                             | 0.01 | 1.00                                                                                | file                                       |
| Length             | 0.5 to 30                                                                              | 0.05 | 10.00                                                                               | all                                        |
| Width              | 0.02 to 4                                                                              | 0.01 | 1.00                                                                                | all but file                               |
| Width x            | 0.1 to 5                                                                               | 0.01 | 1.00                                                                                | file                                       |
| Tail               | 0 to 10                                                                                | 0.05 | 1.00                                                                                | all                                        |
| Tip length         | 0 to 10                                                                                | 0.05 | 1.00                                                                                | pencil · sword · arrow · syringe · breguet |
| Head width         | 0 to 6                                                                                 | 0.05 | 1.20                                                                                | arrow · syringe                            |
| Ring · disc Ø      | 0 to 6                                                                                 | 0.05 | 1.20                                                                                | breguet · lollipop                         |
| Widest at %        | 0 to 100                                                                               | 1    | 75                                                                                  | dauphine                                   |
| Barrel from %      | 0 to 100                                                                               | 1    | 75                                                                                  | syringe                                    |
| Ring · disc at %   | 0 to 100                                                                               | 1    | 75                                                                                  | breguet · lollipop                         |
| Counterweight Ø    | 0 to 4                                                                                 | 0.05 | 0.00                                                                                | all                                        |
| Hollow wall        | 0 to 1                                                                                 | 0.01 | 0.00                                                                                | all                                        |
| Colour             | a colour                                                                               |      |  | all                                        |

| Dial                                  | From · to                    | Step | Starts at                                                                           |
|---------------------------------------|------------------------------|------|-------------------------------------------------------------------------------------|
| Diameter                              | 15 to 50                     | 0.1  | 30.0                                                                                |
| Page margin                           | 0 to 10                      | 0.5  | 3.0                                                                                 |
| Colour                                |                              |      |                                                                                     |
| Plate                                 | a colour                     |      |  |
| Backdrop                              | a colour                     |      |  |
| Date                                  | From · to                    | Step | Starts at                                                                           |
| Date window                           | on · off                     |      | off                                                                                 |
| Position                              | 3 · 4:30 · 6 · 9 · 12        |      |                                                                                     |
| Radius                                | 2 to 18                      | 0.05 | 10.50                                                                               |
| Width                                 | 0.5 to 6                     | 0.05 | 2.60                                                                                |
| Height                                | 0.5 to 5                     | 0.05 | 2.00                                                                                |
| Corner                                | 0 to 2                       | 0.05 | 0.20                                                                                |
| Frame                                 | 0 to 0.6                     | 0.01 | 0.15                                                                                |
| Print clearance                       | 0 to 1                       | 0.01 | 0.25                                                                                |
| Print colour                          | a colour                     |      |  |
| Shows                                 | text                         |      | 17                                                                                  |
| Text % of height                      | 20 to 100                    | 1    | 62                                                                                  |
| Export                                | From · to                    | Step | Starts at                                                                           |
| Production: mono print mask           | on · off                     |      | off                                                                                 |
| Mirror: toner transfer                | on · off                     |      | off                                                                                 |
| Live text in PDF · SVG (fonts need... | on · off                     |      | off                                                                                 |
| PNG dpi                               | 72 to 2400                   | 1    | 600                                                                                 |
| Clip seconds                          | 1 to 60                      | 1    | 4                                                                                   |
| Frames per s                          | 6 to 60                      | 1    | 30                                                                                  |
| Export                                | PDF · SVG · PNG · MP4 · GIF  |      |                                                                                     |
| Settings                              | Save... · Load... · Defaults |      |                                                                                     |

# Find it

You know what you want on the dial. This says what the tool calls it, and where this book shows it.



## Tracks and marks

|                                     |                      |    |
|-------------------------------------|----------------------|----|
| <b>a minute track</b>               | ticks, Count 60      | 6  |
| <b>a railroad track</b>             | two hairline bands   | 25 |
| <b>fifths of a second</b>           | Count 300, skipping  | 25 |
| <b>marks only at 3 · 6 · 9 · 12</b> | Every 3              | 6  |
| <b>all but the cardinals</b>        | Skip those instead   | 6  |
| <b>a scale on an arc</b>            | Start · Span         | 23 |
| <b>a triangle at 12</b>             | 12 o'clock: triangle | 7  |
| <b>a double bar at 12</b>           | 12 o'clock: double   | 7  |
| <b>lume dots</b>                    | Shape dot            | 7  |
| <b>outlined markers</b>             | Hollow wall          | 7  |
| <b>rounded bars</b>                 | Round outer · inner  | 7  |
| <b>my own marker</b>                | Shape file           | 24 |
| <b>ticks that stop at markers</b>   | Skip where           | 8  |
| <b>a gap round the numerals</b>     | Knock out            | 8  |
| <b>a crosshair</b>                  | ticks, Count 4, long | 22 |



## Numerals and colour

|                                 |                     |    |
|---------------------------------|---------------------|----|
| <b>Roman, IIII or IV</b>        | Labels              | 10 |
| <b>a 24-hour ring</b>           | Labels Custom       | 15 |
| <b>05 to 60</b>                 | Labels Numbers      | 18 |
| <b>compass points, words</b>    | Labels Custom       | 10 |
| <b>text on a curve</b>          | Placement on path   | 17 |
| <b>two fonts on one circle</b>  | two Custom lists    | 19 |
| <b>upright or radial</b>        | Placement           | 10 |
| <b>one numeral looks heavy</b>  | Nudge: Size ×       | 11 |
| <b>a variable font's weight</b> | Axes                | 11 |
| <b>Turkish İ i</b>              | Language            | 11 |
| <b>wider spacing</b>            | Tracking            | 11 |
| <b>a chapter ring</b>           | band + rings above  | 9  |
| <b>pie sectors</b>              | Band width = Radius | 9  |
| <b>a red zone</b>               | band, Start · Span  | 23 |
| <b>a hairline circle</b>        | Band width 0.12     | 9  |



## Hands, date, files

|                                   |                   |    |
|-----------------------------------|-------------------|----|
| <b>another hand shape</b>         | Shape             | 12 |
| <b>a lollipop seconds hand</b>    | Ring · disc Ø     | 20 |
| <b>a skeleton hand</b>            | Hollow wall       | 12 |
| <b>my own hand</b>                | Shape file        | 24 |
| <b>ticking or sweeping</b>        | Motion            | 12 |
| <b>another time shown</b>         | Time field        | 3  |
| <b>the date at 4:30</b>           | Position          | 13 |
| <b>no frame round the date</b>    | Frame 0           | 13 |
| <b>a mask for etching</b>         | Production        | 24 |
| <b>flipped for toner transfer</b> | Mirror            | 24 |
| <b>a true-size PDF</b>            | Export → PDF      | 24 |
| <b>a 600 dpi PNG</b>              | PNG dpi           | 24 |
| <b>a moving clip</b>              | MP4 · GIF         | 24 |
| <b>keep or reopen a dial</b>      | Save... · Load... | 2  |
| <b>begin again</b>                | Defaults          | 25 |